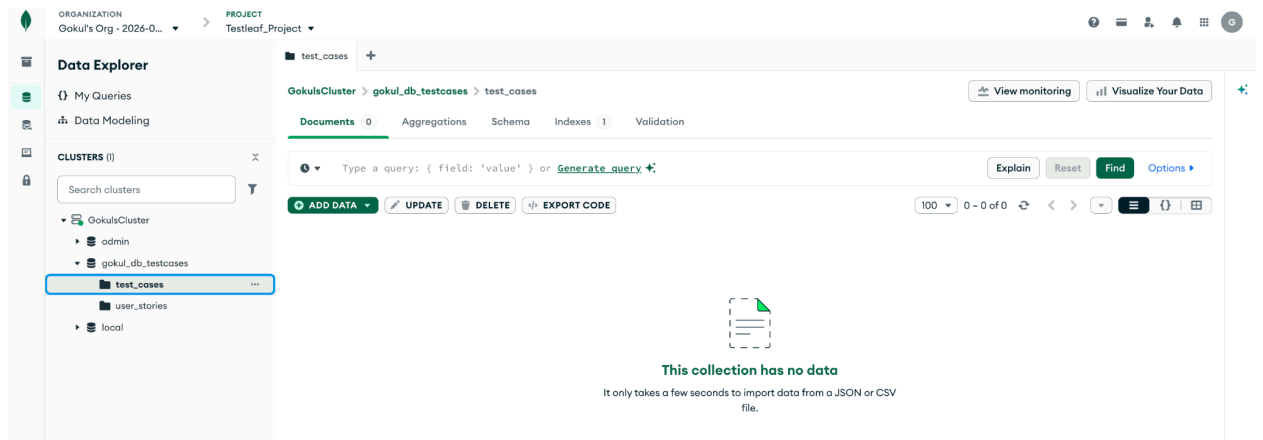
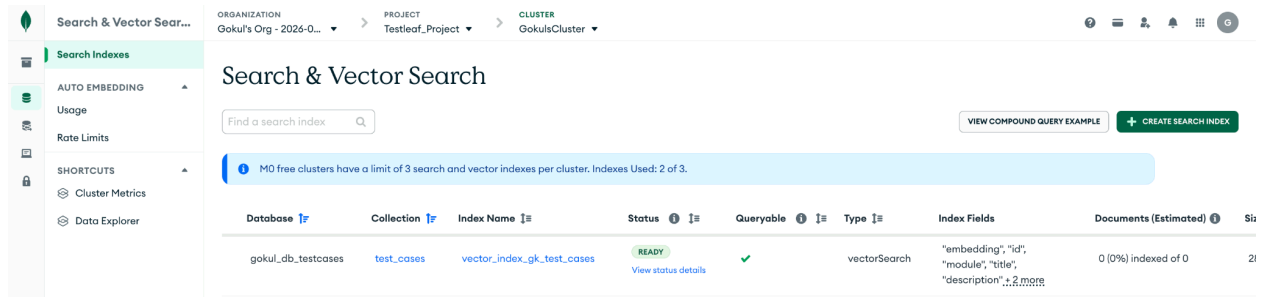


RAG INGESTION BACKEND

A)Test case ingestion via VB studio. Search & Vector search and DB screenshots before ingestion

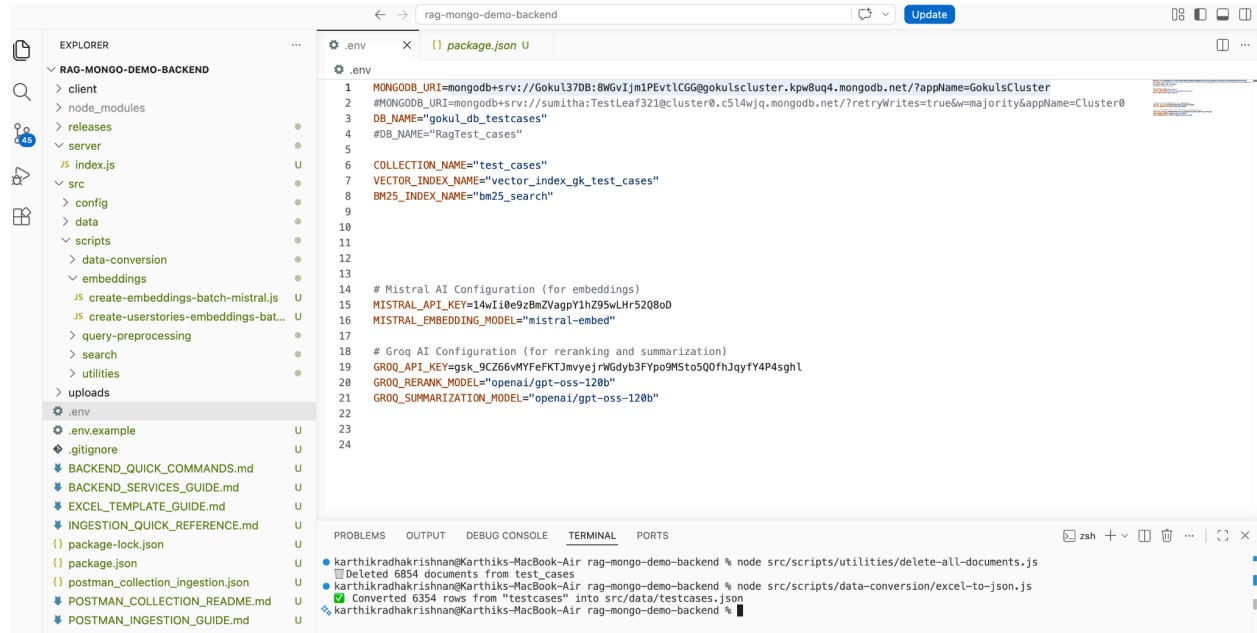


Env file setup



Test case to JSON via command node

src/scripts/data-conversion/[excel-to-json.js](#)



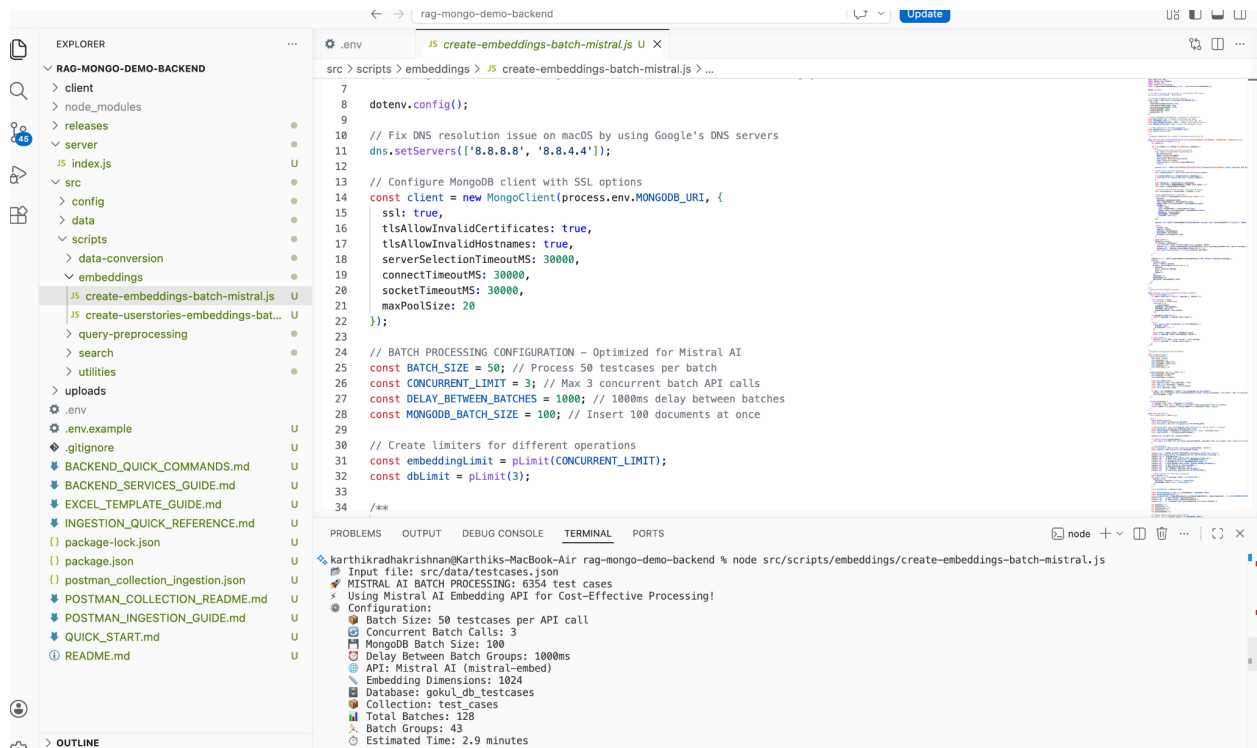
```
.env
1 MONGODB_URI=mongodb+srv://GokuL37DB:8W6vIjmtPEvtLCGG@gokulscuster.kpw8uq4.mongodb.net/?appName=Gokulscuster
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c5l4wj.mongodb.net/?retryWrites=true&w=majority&appName=Cluster0
3 DB_NAME="goku_db_testcases"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="test_cases"
7 VECTOR_INDEX_NAME="vector_index_gk_test_cases"
8 BM25_INDEX_NAME="bm25_search"
9
10
11
12
13
14 # Mistral AI Configuration (for embeddings)
15 MISTRAL_API_KEY=14wIi0e9z8mZVagpY1hZ95wLHr5208oD
16 MISTRAL_EMBEDDING_MODEL="mistral-embed"
17
18 # Groq AI Configuration (for reranking and summarization)
19 GROQ_API_KEY=gsK_9CZ66vMYFeFKTJmvejrwGdyb3FYpo9M5to5Q0fhJqyfY4P4sghL
20 GROQ_RERANK_MODEL="openai/gpt-oss-120b"
21 GROQ_SUMMARIZATION_MODEL="openai/gpt-oss-120b"
22
23
24
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
karthikradhakrishnan@Karthiks-MacBook-Air rag-mongo-demo-backend % node src/scripts/utilities/delete-all-documents.js
Deleted 6854 documents from test_cases
karthikradhakrishnan@Karthiks-MacBook-Air rag-mongo-demo-backend % node src/scripts/data-conversion/excel-to-json.js
Converted 6354 rows from "testcases" into src/data/testcases.json
karthikradhakrishnan@Karthiks-MacBook-Air rag-mongo-demo-backend %
```

Creating embedding via node

src/scripts/embeddings/[create-embeddings-batch-mistral.js](#)



```
.env
src > scripts > embeddings > JS create-embeddings-batch-mistral.js > ...
7
8 dotenv.config();
9
10 // Fix DNS resolution issue on macOS by using Google's DNS servers
11 dns.setServers(['8.8.8.8', '8.8.4.4']);
12
13 // Configure MongoDB client with SSL options
14 const client = new MongoClient(process.env.MONGODB_URI, {
15   ssl: true,
16   tlsAllowInvalidCertificates: true,
17   tlsAllowInvalidHostnames: true,
18   serverSelectionTimeoutMS: 30000,
19   connectTimeoutMS: 30000,
20   socketTimeoutMS: 30000,
21   maxPoolSize: 20
22 });
23
24 // BATCH PROCESSING CONFIGURATION - Optimized for Mistral AI
25 const BATCH_SIZE = 50; // Process 50 testcases per batch
26 const CONCURRENT_LIMIT = 3; // Max 3 concurrent batch API calls
27 const DELAY_BETWEEN_BATCHES = 1000; // 1000ms delay between batches
28 const MONGODB_BATCH_SIZE = 100; // Insert 100 documents at once
29
30 // Create limiters for different operations
31 const embeddingLimit = pLimit(CONCURRENT_LIMIT);
32 const dbLimit = pLimit(3);
33
34 /**
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
node src/scripts/embeddings/create-embeddings-batch-mistral.js
Input file: src/data/testcases.json
MISTRAL AI BATCH PROCESSING: 6354 test cases
Using Mistral AI Embedding API for Cost-Effective Processing!
Configuration:
Batch Size: 50 testcases per API call
Concurrent Batch Calls: 3
MongoDB Batch Size: 100
Delay Between Batch Groups: 1000ms
API: Mistral AI (mistral-embed)
Embedding Dimensions: 1024
Database: gokul_db_testcases
Collection: test_cases
Total Batches: 128
Batch Groups: 43
Estimated Time: 2.9 minutes
```

Create Embedding Completed

```
connectTimeoutMS: 30000,
socketTimeoutMS: 30000,
maxPoolSize: 20
});

// BATCH PROCESSING CONFIGURATION - optimized for Mistral AI

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[Batch 118/128] Success! Cost: $0.000787 | Tokens: 7873
Waiting 1000ms before next batch group...
[Batch 121/128] Processing 50 testcases with Mistral AI...
[Batch 122/128] Processing 50 testcases with Mistral AI...
[Batch 123/128] Processing 50 testcases with Mistral AI...
[Batch 121/128] Success! Cost: $0.000841 | Tokens: 8413
[Batch 122/128] Success! Cost: $0.000882 | Tokens: 8820
[Batch 123/128] Success! Cost: $0.000852 | Tokens: 8520
Progress: 6050/6354 (95.2%) | Rate: 40.1/sec | ETA: 8s | Cost: $0.112091
Waiting 1000ms before next batch group...
[Batch 124/128] Processing 50 testcases with Mistral AI...
[Batch 125/128] Processing 50 testcases with Mistral AI...
[Batch 126/128] Success! Cost: $0.000946 | Tokens: 9461
[Batch 125/128] Success! Cost: $0.001018 | Tokens: 10183
[Batch 124/128] Success! Cost: $0.000846 | Tokens: 8458
Waiting 1000ms before next batch group...
[Batch 127/128] Processing 50 testcases with Mistral AI...
[Batch 128/128] Processing 4 testcases with Mistral AI...
[Batch 128/128] Success! Cost: $0.000074 | Tokens: 744
[Batch 127/128] Success! Cost: $0.001032 | Tokens: 10317
Progress: 6354/6354 (100.0%) | Rate: 40.3/sec | ETA: 0s | Cost: $0.117741

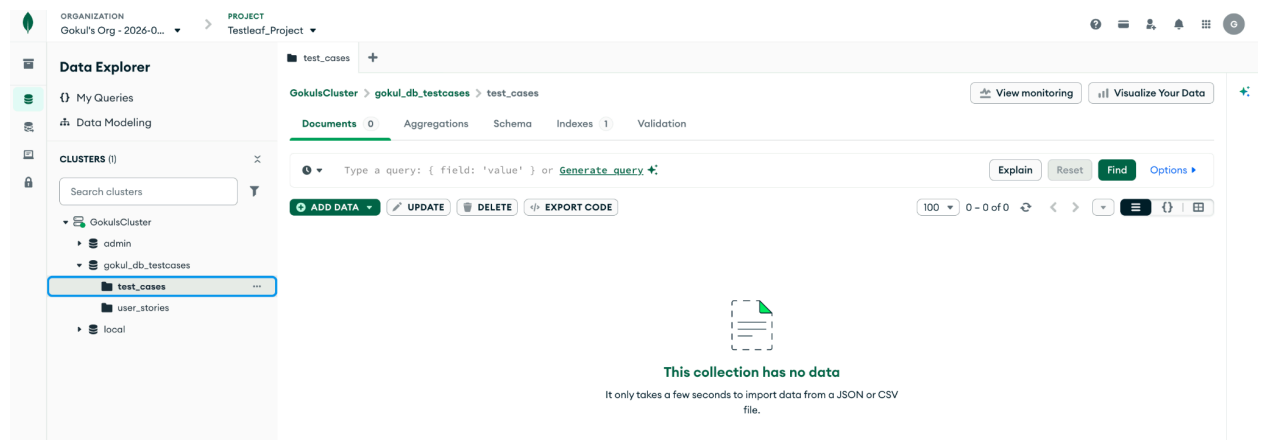
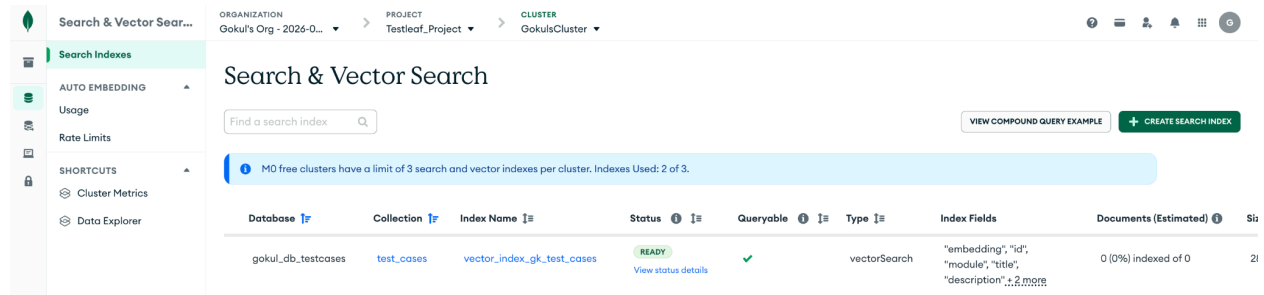
MISTRAL AI BATCH PROCESSING COMPLETE!
Final Statistics:
  Total Time: 2m 38s
  Processing Rate: 40.1 testcases/second
  Total Test Cases: 6354
  Successfully Processed: 6354
  Failed: 0
```

Search & Vector search and DB screenshots after ingestion

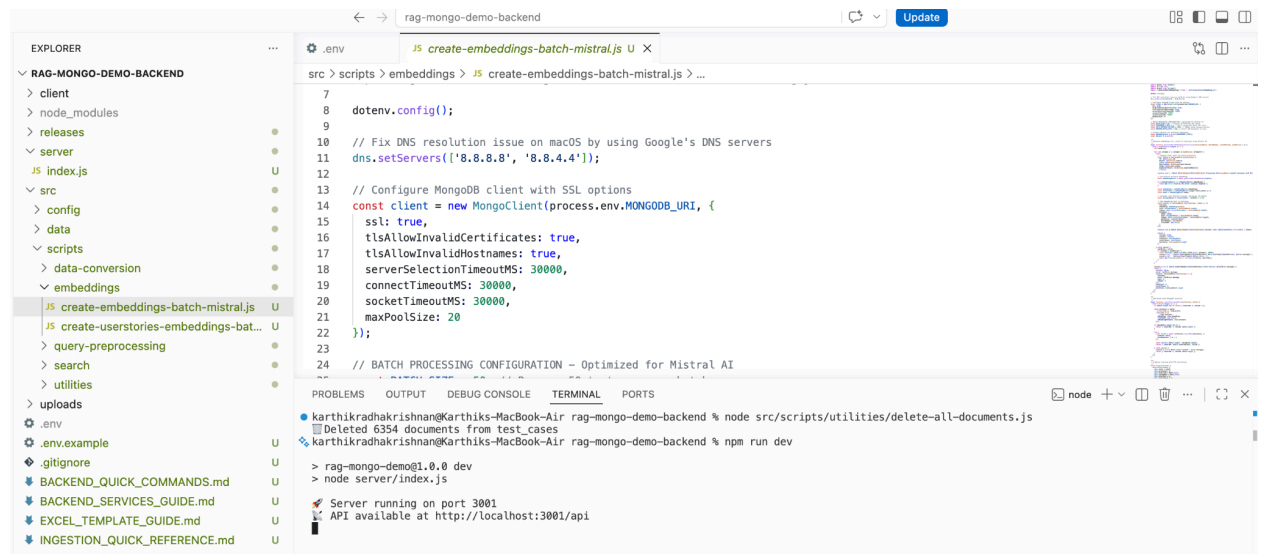
Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
gokul_db_testcases	test_cases	vector_index_gk_test_cases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", ... 2 more	6,354 (100%) Indexed of 6,354	26.47M

_id	module	id	preRequisites	title	description	steps	expectedResults	automationManual
ObjectID('6a06128e57da8a52c8e225c4')	"Registration"	"TC_005"	"PRN/ERN Number"	"Conversion from PRN/ERN Reg to UHID"	"Verify Conversion from PRN/ERN Reg to UHID"	1. Launch the application 2. Login to the application with valid creden."	"Should get convert successfully from PRN/ERN Reg to UHID"	"Automated"

A) Test case ingestion via VB studio. Search & Vector search and DB screenshots before ingestion



Npm run dev to make the server running



Health check from postman

The screenshot shows the Postman interface for a GET request to `{{baseUrl}}/api/health`. The left sidebar lists collections, with 'RAG MongoDB - Data Ingestion' expanded to show 'GET 1. Health Check'. The main area shows the request details and a successful response with status 200 OK, 16 ms, and 312 B. The response body is a JSON object:

```
{
  "status": "OK",
  "message": "Server is running"
}
```

Test Case to JSON from postman

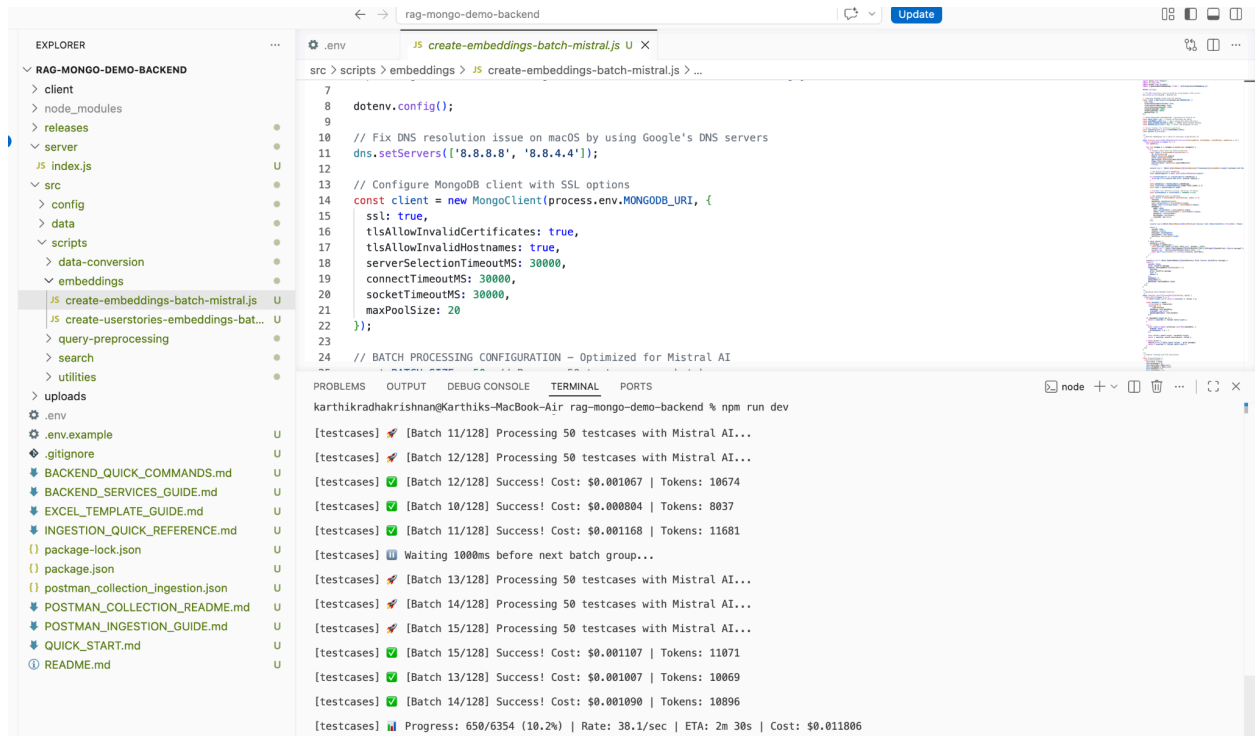
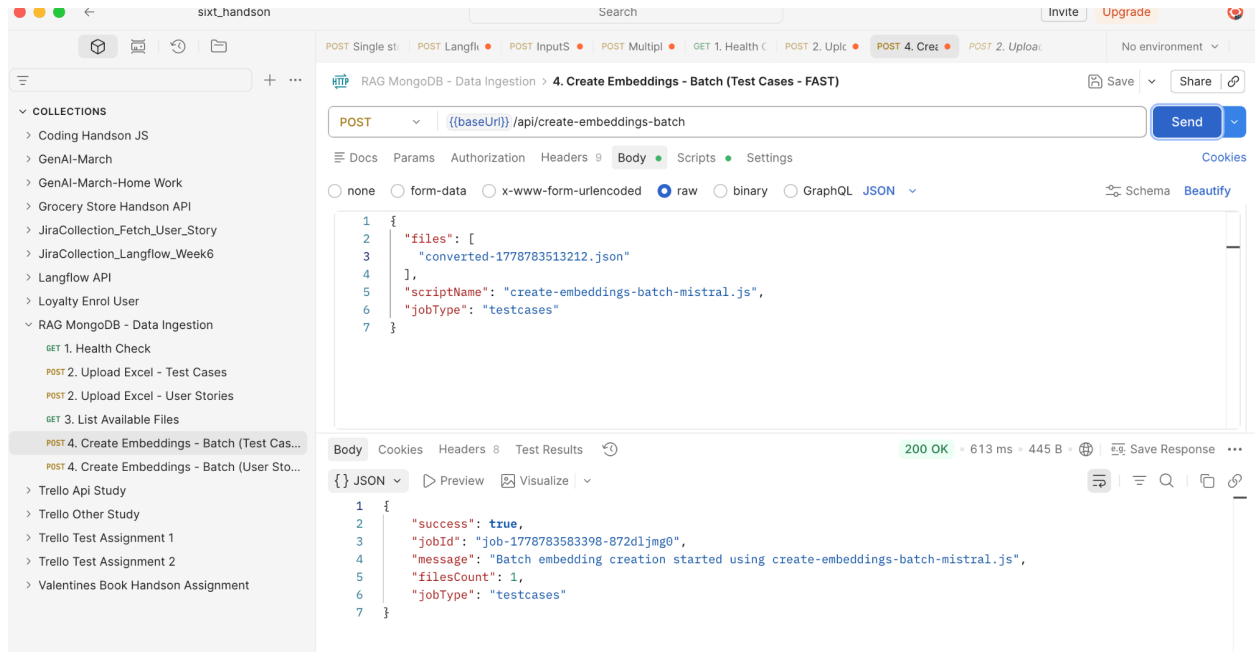
The screenshot shows the Postman interface for a POST request to `{{baseUrl}}/api/upload-excel`. The left sidebar lists collections, with 'RAG MongoDB - Data Ingestion' expanded to show 'POST 2. Upload Excel - Test Cases'. The main area shows the request details and a successful response with status 200 OK, 346 ms, and 627 B. The response body is a JSON object:

```
{
  "success": true,
  "message": "Test cases file converted successfully",
  "outputFile": "converted-1778783513212.json",
  "output": "⚠ Sheet 'Testcases' not found. Using first sheet: 'testcases'\n✅ Converted 6354 rows from 'testcases' into /Users/karthikzradhakrishnan/Downloads/Goku1/Week7/D1/rag-mongo-demo-backend/src/data/converted-1778783513212.json\n"
```

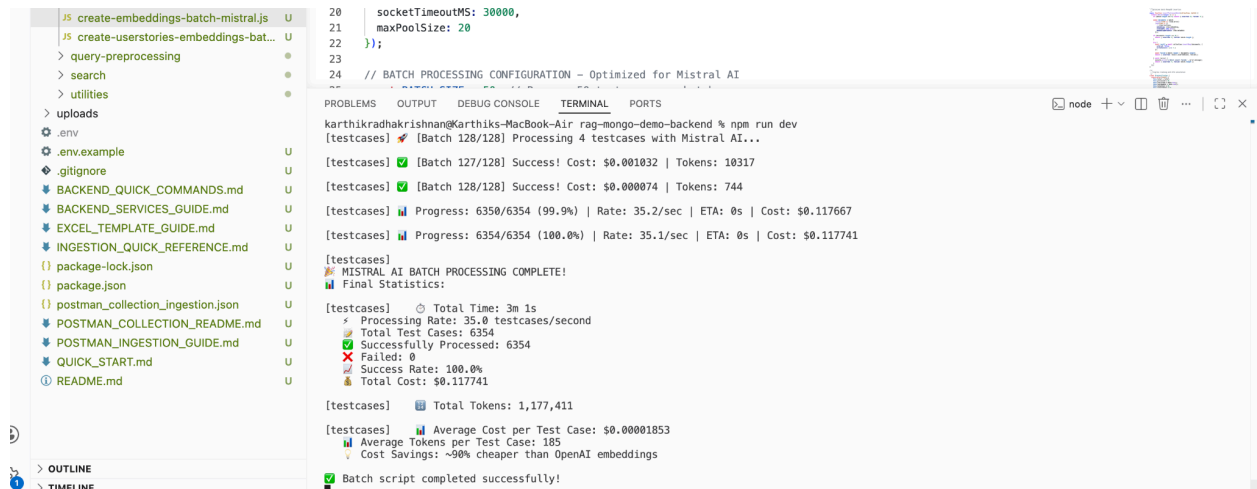
Below is a table representing the data from the 'testcases.xlsx' file as shown in the request body tab:

Key	Value	Description	Bulk Edit
file	testcases.xlsx	Excel file containing test cases	
dataType	testcases	Type of data: 'testcases' or 'userstories'	
sheetName	Testcases	Name of the Excel sheet to convert (optional, d	
Key	Value	Description	

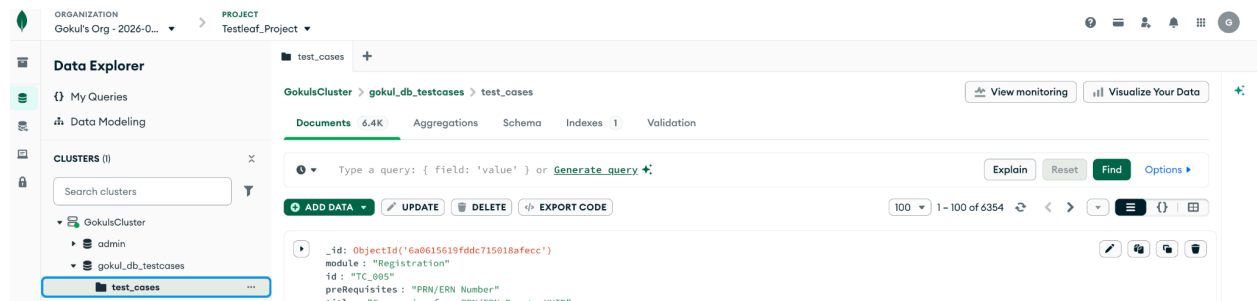
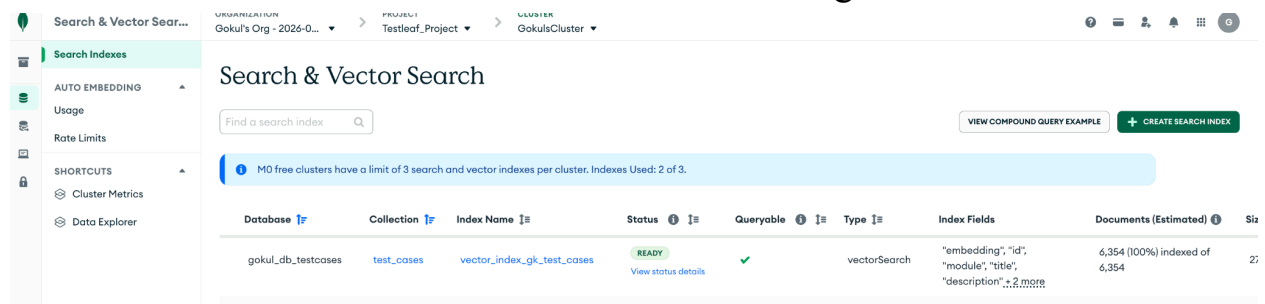
Creating embedding via postman and batch started in VB studio



Create Embedding Completed

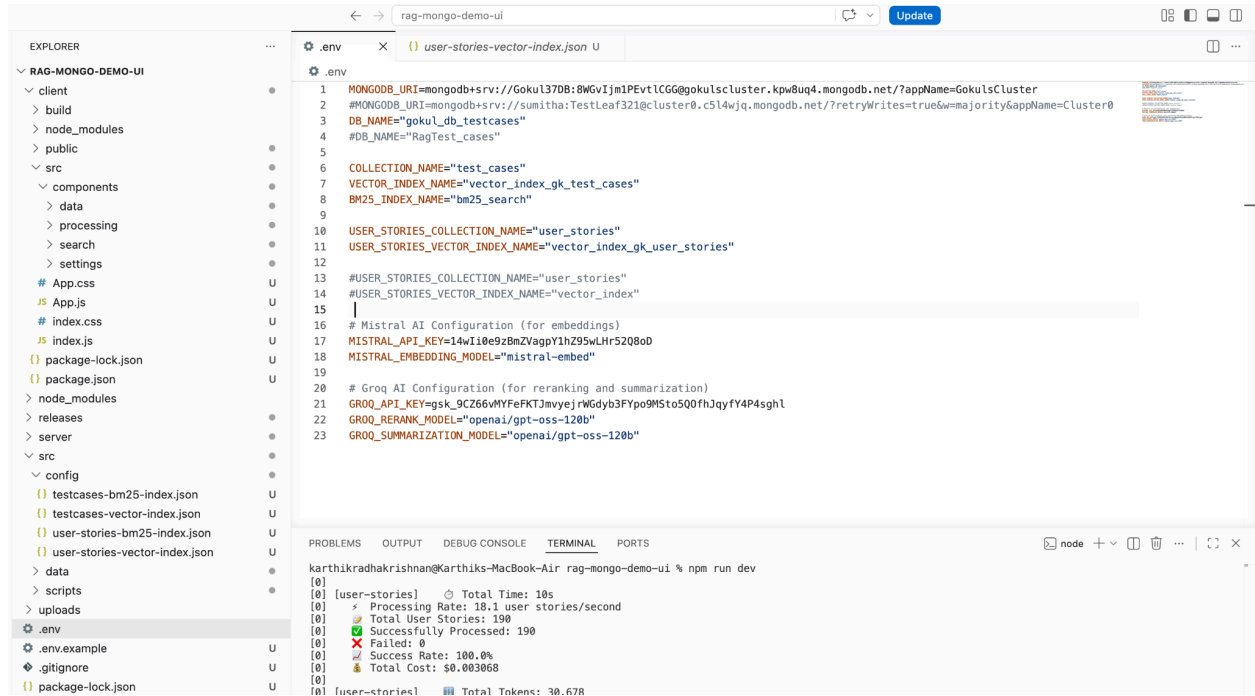


Search & Vector search and DB screenshots after ingestion

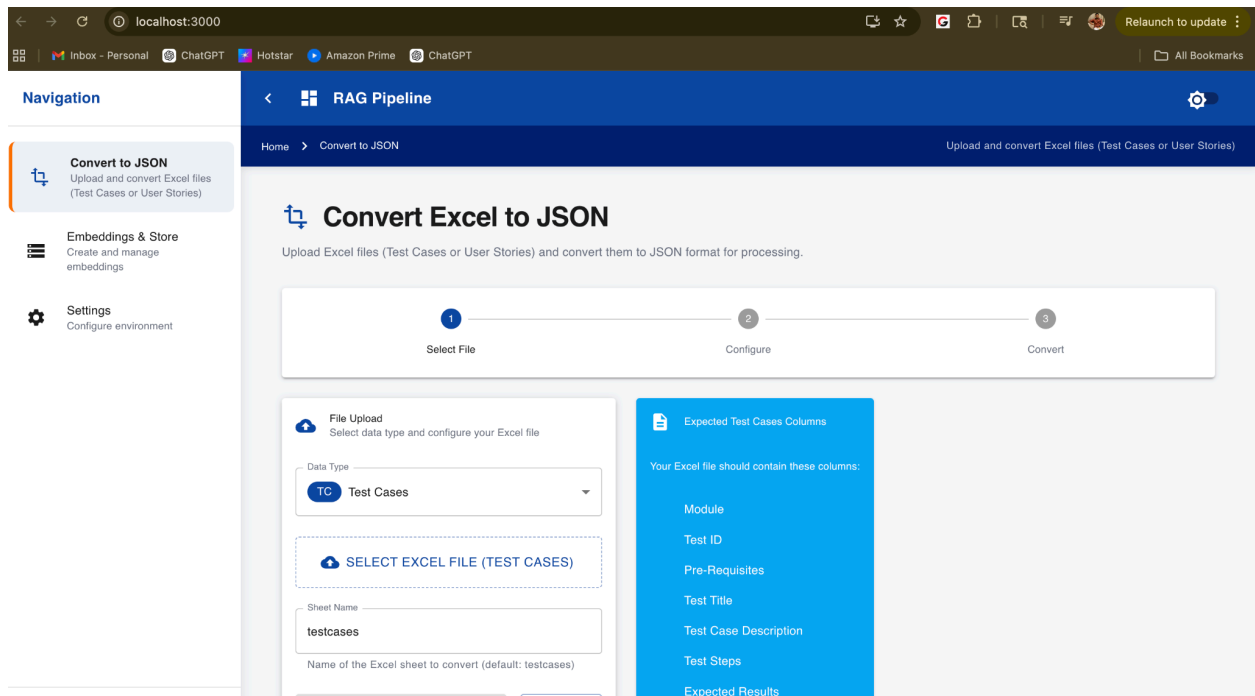


RAG INGESTION FRONTEND

Env file updated for test case and user story



UI host opened by running npm run dev



Converting test cases to JSON, creating embedding via UI

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Settings

Configure environment

RAG Pipeline

Home > Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Convert Excel to JSON

Upload Excel files (Test Cases or User Stories) and convert them to JSON format for processing.

1

Select File

2

Configure

3

Convert

File Upload

Select data type and configure your Excel file

Data Type

TC Test Cases

SELECT EXCEL FILE (TEST CASES)

testcases.xlsx

917.3 KB

Selected

Sheet Name

testcases

Expected Test Cases Columns

Your Excel file should contain these columns:

Module

Test ID

Pre-Requisites

Test Title

Test Case Description

Test Steps

Expected Results

Automation/Manual

File selected successfully

RAG Demo v1.2

Enterprise Edition

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Settings

Configure environment

Home > Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

testcases

Name of the Excel sheet to convert (default: testcases)

CONVERT TEST CASES

RESET

Automation/Manual

Priority

Created By

Created Date

Last modified date

Risk

Version

Type

Conversion Results

Success! Test cases file converted successfully

Output File

converted-1778784416179.json

Saved to data folder

Status

Completed

Ready for embedding creation

Conversion Log

RAG Demo v1.2

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Settings

Configure environment

RAG Pipeline

Create and manage embeddings

Home > Embeddings & Store

File Name	Size	Uploaded	Type
☐ converted-1778779156705.json	540.19 KB	14/05/2026 22:49:16	JSON
☒ converted-1778784416179.json	6.29 MB	15/05/2026 00:16:56	JSON
☐ stories-1770980345846.json	198.29 KB	13/02/2026 16:29:07	JSON
☐ stories-1770988242140.json	198.29 KB	13/02/2026 18:40:44	JSON

1 row selected Rows per page: 10 1-10 of 12

Selected 1 file(s): converted-1778784416179.json

Actions

Process selected files

CREATE EMBEDDINGS

index.js

package-lock.json

package.json

node_modules

releases

server

src

config

testcases-bm25-index.json

testcases-vector-index.json

user-stories-bm25-index.json

user-stories-vector-index.json

data

scripts

uploads

.env

.env.example

.gitignore

package-lock.json

package.json

23

GROQ_SUMMARIZATION_MODEL="openai/gpt-oss-120b"

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

karthikradhakrishnan@Karthiks-MacBook-Air rag-mongo-demo-ui % npm run dev

[0] [testcases] [Waiting 1000ms before next batch group...]

[0] [testcases] [Batch 7/128] Processing 50 testcases with Mistral AI...

[0] [testcases] [Batch 8/128] Processing 50 testcases with Mistral AI...

[0] [testcases] [Batch 9/128] Processing 50 testcases with Mistral AI...

[0] [testcases] [Batch 8/128] Success! Cost: \$0.000992 | Tokens: 9920

[0] [testcases] [Batch 9/128] Success! Cost: \$0.000896 | Tokens: 8955

[0] [testcases] [Batch 7/128] Success! Cost: \$0.000925 | Tokens: 9248

[0] [testcases] [Waiting 1000ms before next batch group...]

[0] [testcases] [Batch 10/128] Processing 50 testcases with Mistral AI...

[0] [testcases] [Batch 11/128] Processing 50 testcases with Mistral AI...

[0] [testcases] [Batch 12/128] Processing 50 testcases with Mistral AI...

[0] [testcases] [Batch 11/128] Success! Cost: \$0.001168 | Tokens: 11681

[0] [testcases] [Batch 12/128] Success! Cost: \$0.001067 | Tokens: 10674

[0] [testcases] [Batch 10/128] Success! Cost: \$0.000804 | Tokens: 8037

Batch successful:

config

testcases-bm25-index.json

testcases-vector-index.json

user-stories-bm25-index.json

user-stories-vector-index.json

data

scripts

uploads

.env

.env.example

.gitignore

package-lock.json

package.json

[0] [testcases] [Batch 128/128] Success! Cost: \$0.000074 | Tokens: 744

[0] [testcases] [Batch 127/128] Success! Cost: \$0.001032 | Tokens: 10317

[0] [testcases] Progress: 6354/6354 (100.0%) | Rate: 43.4/sec | ETA: 0s | Cost: \$0.117741

[0] [testcases] MISTRAL AI BATCH PROCESSING COMPLETE!

[0] Final Statistics:

[0] Total Time: 2m 27s

[0] Processing Rate: 43.2 testcases/second

[0] Total Test Cases: 6354

[0] Successfully Processed: 6354

[0] Failed: 0

[0] Success Rate: 100.0%

[0] Total Cost: \$0.117741

[0] Total Tokens: 1,177,411

[0] Average Cost per Test Case: \$0.00001853

[0] Average Tokens per Test Case: 185

[0] Cost Savings: ~90% cheaper than OpenAI embeddings

[0] Batch script completed successfully!

Search & Vector search and DB screenshots after ingestion from UI

The top screenshot shows the 'Search & Vector Search' interface. It includes a sidebar with 'Search Indexes', 'AUTO EMBEDDING', 'Usage', 'Rate Limits', 'SHORTCUTS', 'Cluster Metrics', and 'Data Explorer'. The main area displays a table of search indexes:

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
gokul_db_testcases	test_cases	vector_index_gk_test_cases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description"...2 more	6,354 (100%) indexed of 6,354	13

Below the table, a document is displayed in the 'Data Explorer' view. The document is a JSON object with fields like 'id', 'module', 'preRequisites', 'title', 'description', 'steps', 'expectedResults', 'automationManual', 'priority', 'createdBy', 'createdDate', 'lastModifiedDate', 'linkedStories', 'risk', 'version', 'type', 'embedding', 'createdAt', and 'embeddingMetadata'.

Additional cases added:

The bottom screenshot shows the 'Data Explorer' interface. The left sidebar lists clusters: 'GokulsCluster', 'admin', 'gokul_db_testcases', 'test_cases', 'user_stories', and 'local'. The 'test_cases' cluster is selected. The main area displays a document with fields like 'id', 'module', 'preRequisites', 'title', 'description', 'steps', 'expectedResults', 'automationManual', 'priority', 'createdBy', 'createdDate', 'lastModifiedDate', 'linkedStories', 'risk', 'version', 'type', 'embedding', 'createdAt', and 'embeddingMetadata'.

UI for user story (practise):

Navigation

**Convert to JSON**
Upload and convert Excel files
(Test Cases or User Stories)

**Embeddings & Store**
Create and manage
embeddings

**Settings**
Configure environment

<  RAG Pipeline 

Home > Convert to JSON Upload and convert Excel files (Test Cases or User Stories)

Upload Excel files (Test Cases or User Stories) and convert them to JSON format for processing.

1 Select File

2 Configure

3 Convert

**File Upload**
Select data type and configure your Excel file

Data Type

US

User Stories

 SELECT EXCEL FILE (USER STORIES)

Sheet Name

stories

Name of the Excel sheet to convert (default: stories)

 CONVERT USER STORIES

RESET

**Expected User Stories Columns**

Your Excel file should contain these columns:

Story ID / Key

Summary / Title

Description

Status

Priority

Assignee

Reporter

Story Points

Components

User story to JSON:

Navigation

**Convert to JSON**
Upload and convert Excel files
(Test Cases or User Stories)

**Embeddings & Store**
Create and manage
embeddings

**Settings**
Configure environment

<  RAG Pipeline 

Home > Convert to JSON Upload and convert Excel files (Test Cases or User Stories)

Select File

Configure

Convert

**File Upload**
Select data type and configure your Excel file

Data Type

US

User Stories

 SELECT EXCEL FILE (USER STORIES)

 **userstories.xlsx**
25.6 KB

Selected

Sheet Name

stories

Name of the Excel sheet to convert (default: stories)

 CONVERT USER STORIES

RESET

**Expected User Stories Columns**

Your Excel file should contain these columns:

Story ID / Key

Summary / Title

Description

Status

Priority

Assignee

Reporter

Story Points

Components

Labels

Sprint

Epic

Acceptance Criteria

Created Date

RAG Demo v1.2
Enterprise Edition

 User Stories converted successfully!

Created Date

Updated Date

✓ Conversion Results

✓ Success! User stories file converted successfully

Output File

stories-1778785423798.json

Saved to data folder

Status

Completed

Ready for embedding creation

Conversion Log

📄 Converting 190 rows from sheet "stories"...

✓ Converted 190 user stories from "stories" into /Users/karthikradhakrishnan/Downloads/Goku1/Week7/D2/rag-mongo-demo-ui/src/data/stories-1778785423798.json

⚠ Filtered out: 0 empty rows

Available Files

 REFRESH

14 JSON files found

	File Name	Size	Modified	Type
☐	 stories-1770988242140.json	198.29 KB	13/02/2026 18:40:44	JSON
☐	 stories-1770988363821.json	198.29 KB	13/02/2026 18:42:45	JSON
☐	 stories-1778780312973.json	198.29 KB	14/05/2026 23:08:33	JSON
☒	 stories-1778785423798.json	198.29 KB	15/05/2026 00:33:43	JSON

The screenshot displays a terminal window with a file explorer sidebar on the left. The sidebar shows a project structure with folders like `node\_modules`, `releases`, `server`, `src`, `config`, `data`, `uploads`, and `env`. The main terminal area shows the output of a command to run a batch script. The output indicates that 190 user stories were processed successfully at a rate of 40.3/sec, with a total cost of \$0.003068. It also shows final statistics such as a total time of 5 seconds and an average cost per user story of \$0.00001615.

```
karthikradhakrishnan@Karthiks-MacBook-Air rag-mongo-demo-ui % npm run dev
[0] [user-stories] [Batch 2/4] Success! Cost: $0.000008 | Tokens: 8077
[0] 
[0] [user-stories] [Batch 3/4] Success! Cost: $0.000008 | Tokens: 8077
[0] [user-stories] [Batch 4/4] Processing 40 user stories with Mistral AI...
[0] 
[0] [user-stories] [Batch 4/4] Success! Cost: $0.000641 | Tokens: 6413
[0] 
[0] [user-stories] Progress: 190/190 (100.0%) | Rate: 40.3/sec | ETA: 0s | Cost: $0.003068
[0] 
[0] [user-stories] MISTRAL AI BATCH PROCESSING COMPLETE!
[0] 
[0] [user-stories] Final Statistics:
[0]   Total Time: 5s
[0]   Processing Rate: 34.7 user stories/second
[0]   Total User Stories: 190
[0]   Successfully Processed: 190
[0]   Failed: 0
[0]   Success Rate: 100.0%
[0]   Total Cost: $0.003068
[0] 
[0] [user-stories] Total Tokens: 30,678
[0] 
[0] [user-stories] Average Cost per User Story: $0.00001615
[0]   Average Tokens per User Story: 161
[0]   Cost Savings: ~90% cheaper than OpenAI embeddings
[0] 
[0] Batch script completed successfully!
```

Search & Vector search and DB screenshots for User story after ingestion from UI

Search & Vector Search

Find a search index

[VIEW COMPOUND QUERY EXAMPLE](#) [+ CREATE SEARCH INDEX](#)

1 M0 free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 2 of 3.

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
gokul_db_testcases	test_cases	vector_index_gk_test_cases	READY View status details	✓	vectorSearch	"embedding", "id", "module", "title", "description", "tags"	0 (0%) indexed of 0	3
gokul_db_testcases	user_stories	vector_index_gk_user_stories	READY View status details	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "tags"	190 (100%) indexed of 190	3

[Learn more about Atlas Search](#)

Organization: Gokul's Org - 2026-0... **Project:** Testleaf_Project

Data Explorer

- My Queries
- Data Modeling

CLUSTERS (1)

- GokulsCluster
 - admin
 - gokul_db_testcases
 - test_cases
 - user_stories**
 - local

user_stories

GokulsCluster > gokul_db_testcases > user_stories

[View monitoring](#) [Visualize Your Data](#)

Documents 190 | Aggregations | Schema | Indexes | Validation

Type a query: { field: 'value' } or [Generate query](#)

[EXPLAIN](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

100 1 - 100 of 190

```
{
  "_id": ObjectId("6a061d03c3ae642ed3a4b1f5"),
  "key": "HC-251",
  "summary": "Consultant Dashboard My Inpatients",
  "description": "As a user, I want to consultant dashboard my inpatients so that the re...",
  "status": Object,
  "priority": Object,
  "assignee": null,
  "reporter": null,
  "created": "2022-01-06T00:00:00.000Z",
  "updated": "2026-05-14T19:03:43.986Z",
  "components": Array (empty),
  "labels": Array (empty),
  "fixVersions": Array (empty),
  "storyPoints": null,
  "project": "Health Care",
  "epic": "Emergency Management",
  "acceptanceCriteria": "Given a user with valid access, when they open the dashboard, then rel...",
  "businessValue": "",
  "risk": "Critical",
  "dependencies": "",
  "notes": "",
  "sprint": null,
  "team": null,
  "issueLinks": Array (empty),
  "url": ""
}
```

[Show 6 more fields](#)